

## TREASURE COAST INTL, FL, USA

WMO: 722103

Lat: 27.498N

Lon: 80.377W

Elev: 24

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 12895

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 38.1       | 42.7 | 26.7                          | 20.7 | 45.5 | 32.7 | 27.2 | 50.1 | 21.6                  | 70.8 | 19.3 | 71.9 | 4.4                   | 310  | 0.357 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 15.3                   | 91.6            | 77.4 | 90.4 | 77.6 | 89.4 | 77.8 | 80.4                | 87.5 | 79.8 | 87.1 | 79.1 | 86.4 | 9.7                  | 110  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.6                            | 148.4 | 83.9 | 77.5 | 143.0 | 83.6 | 77.0 | 140.6 | 83.2 | 43.7          | 87.5 | 43.0 | 86.6 | 42.4 | 86.7 | 84.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 19.7              | 17.8 | 16.3 | DB                         | 32.5 | 95.5               | 3.5 | 1.8                                                | 30.0 | 96.8       | 28.0 | 97.9       | 26.0 | 98.9       | 23.5 | 100.2 |
|                   |      |      | WB                         | 30.1 | 81.6               | 3.3 | 0.9                                                | 27.7 | 82.2       | 25.8 | 82.8       | 23.9 | 83.3       | 21.5 | 84.0  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 73.3   | 61.8  | 65.1  | 67.7  | 71.9  | 76.4  | 80.1  | 81.4  | 81.7  | 80.3  | 76.3  | 69.8  | 66.1 |
|                                                                                  | DBStd       | 8.57   | 8.06  | 7.39  | 6.66  | 4.96  | 3.47  | 2.20  | 1.87  | 1.64  | 2.04  | 4.79  | 6.07  | 7.81 |
|                                                                                  | HDD50       | 19     | 11    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5    |
|                                                                                  | HDD65       | 417    | 156   | 83    | 52    | 9     | 0     | 0     | 0     | 0     | 0     | 4     | 28    | 85   |
|                                                                                  | CDD50       | 8511   | 378   | 425   | 550   | 656   | 819   | 904   | 972   | 983   | 910   | 816   | 593   | 505  |
|                                                                                  | CDD65       | 3434   | 58    | 85    | 137   | 215   | 354   | 454   | 507   | 518   | 460   | 355   | 171   | 120  |
|                                                                                  | CDH74       | 30157  | 297   | 498   | 838   | 1615  | 3110  | 4383  | 5176  | 5306  | 4385  | 2984  | 1007  | 558  |
|                                                                                  | CDH80       | 9765   | 23    | 74    | 155   | 332   | 887   | 1637  | 2127  | 2162  | 1503  | 694   | 117   | 54   |
| Wind                                                                             | WSAvg       | 6.8    | 6.7   | 6.9   | 7.9   | 8.2   | 8.1   | 6.4   | 5.6   | 5.7   | 5.9   | 7.1   | 6.7   | 6.7  |
| Precipitation                                                                    | PrecAvg     | 53.5   | 2.3   | 3.2   | 3.2   | 2.6   | 4.3   | 6.1   | 5.9   | 6.5   | 8.2   | 6.0   | 3.2   | 2.2  |
|                                                                                  | PrecMax     | 80.2   | 9.4   | 9.7   | 12.4  | 10.4  | 13.0  | 15.8  | 13.1  | 17.7  | 19.9  | 14.3  | 9.3   | 8.2  |
|                                                                                  | PrecMin     | 32.7   | 0.2   | 0.1   | 0.1   | 0.2   | 0.4   | 0.8   | 0.8   | 1.3   | 1.5   | 0.7   | 0.3   | 0.1  |
|                                                                                  | PrecStd     | 10.9   | 1.8   | 2.3   | 2.6   | 2.0   | 2.5   | 3.0   | 3.0   | 3.3   | 4.4   | 3.5   | 2.4   | 2.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 82.5  | 85.6  | 88.1  | 89.2  | 92.3  | 93.2  | 93.4  | 93.3  | 91.4  | 89.4  | 85.7  | 84.2 |
|                                                                                  |             | MCWB   | 70.3  | 70.4  | 72.1  | 72.2  | 74.2  | 77.0  | 77.6  | 77.7  | 77.3  | 76.4  | 74.9  | 73.4 |
|                                                                                  | 2%          | DB     | 79.9  | 82.2  | 84.1  | 86.1  | 89.0  | 90.6  | 91.2  | 91.1  | 89.5  | 87.0  | 82.9  | 81.3 |
|                                                                                  |             | MCWB   | 69.6  | 70.5  | 70.8  | 72.5  | 74.3  | 77.3  | 78.1  | 78.4  | 78.2  | 76.5  | 73.4  | 72.1 |
|                                                                                  | 5%          | DB     | 77.4  | 79.5  | 81.2  | 83.6  | 86.4  | 89.2  | 90.0  | 90.0  | 88.1  | 85.3  | 81.2  | 79.3 |
|                                                                                  |             | MCWB   | 68.4  | 69.7  | 70.3  | 71.9  | 74.2  | 77.3  | 78.2  | 78.6  | 77.7  | 75.9  | 72.1  | 71.0 |
|                                                                                  | 10%         | DB     | 74.9  | 77.0  | 78.8  | 81.5  | 84.4  | 87.7  | 88.5  | 88.6  | 86.8  | 83.8  | 79.3  | 77.2 |
|                                                                                  |             | MCWB   | 67.1  | 68.2  | 68.6  | 70.4  | 73.6  | 77.0  | 78.0  | 78.3  | 77.2  | 74.8  | 70.2  | 69.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 73.1  | 73.5  | 75.3  | 76.4  | 78.1  | 80.4  | 81.3  | 81.3  | 80.6  | 79.9  | 76.8  | 75.3 |
|                                                                                  |             | MCDB   | 78.6  | 79.8  | 81.3  | 84.6  | 85.4  | 88.1  | 88.0  | 88.6  | 86.9  | 84.9  | 82.2  | 80.5 |
|                                                                                  | 2%          | WB     | 71.3  | 72.4  | 73.4  | 74.9  | 76.8  | 79.3  | 80.3  | 80.3  | 79.8  | 78.3  | 75.4  | 73.9 |
|                                                                                  |             | MCDB   | 76.6  | 78.6  | 80.0  | 82.2  | 84.5  | 87.0  | 87.4  | 87.8  | 86.5  | 83.6  | 80.5  | 79.1 |
|                                                                                  | 5%          | WB     | 70.1  | 71.1  | 72.0  | 73.5  | 75.9  | 78.4  | 79.5  | 79.6  | 78.9  | 77.4  | 74.1  | 72.8 |
|                                                                                  |             | MCDB   | 75.3  | 77.1  | 78.7  | 80.5  | 83.4  | 85.9  | 87.0  | 87.3  | 85.5  | 83.1  | 79.2  | 77.7 |
|                                                                                  | 10%         | WB     | 68.8  | 69.7  | 70.3  | 72.4  | 75.0  | 77.8  | 78.6  | 78.9  | 78.2  | 76.4  | 72.6  | 71.3 |
|                                                                                  |             | MCDB   | 74.0  | 75.5  | 76.7  | 79.2  | 82.3  | 85.1  | 86.1  | 86.4  | 84.6  | 81.9  | 77.7  | 75.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.6  | 19.9  | 19.7  | 18.6  | 16.6  | 15.5  | 15.6  | 15.3  | 13.8  | 14.4  | 17.2  | 17.8 |
|                                                                                  |             | MCDBR  | 19.7  | 20.1  | 20.6  | 20.1  | 19.6  | 17.6  | 17.2  | 16.6  | 15.3  | 14.4  | 16.1  | 16.9 |
|                                                                                  |             | MCWBR  | 11.4  | 10.5  | 10.0  | 9.6   | 8.3   | 7.3   | 6.9   | 6.8   | 6.4   | 6.5   | 8.5   | 9.6  |
|                                                                                  | 5% WB       | MCDBR  | 17.8  | 17.9  | 18.6  | 17.8  | 16.8  | 15.8  | 15.6  | 15.2  | 13.9  | 12.4  | 14.2  | 15.3 |
| MCWBR                                                                            |             | 11.2   | 9.8   | 10.1  | 9.0   | 7.7   | 7.2   | 7.0   | 6.8   | 6.4   | 6.0   | 8.0   | 9.4   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.348  | 0.356 | 0.371 | 0.396 | 0.426 | 0.452 | 0.469 | 0.458 | 0.425 | 0.395 | 0.364 | 0.359 |      |
|                                                                                  | taud        | 2.531  | 2.527 | 2.470 | 2.378 | 2.338 | 2.339 | 2.286 | 2.340 | 2.426 | 2.476 | 2.510 | 2.522 |      |
|                                                                                  | Ebn at Noon | 285    | 292   | 292   | 286   | 276   | 267   | 262   | 265   | 272   | 275   | 277   | 276   |      |
|                                                                                  | Edh at Noon | 29     | 31    | 35    | 39    | 41    | 40    | 43    | 40    | 36    | 32    | 29    | 28    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1099   | 1354  | 1652  | 1928  | 1978  | 1858  | 1834  | 1773  | 1562  | 1403  | 1157  | 998   |      |
|                                                                                  | RadStd      | 77     | 75    | 96    | 68    | 150   | 158   | 122   | 110   | 79    | 91    | 56    | 69    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            |       | +0.92   | N/A    | +3.28   | N/A   | N/A   | N/A         | N/A   | +297  | +272  |
| (47) Regional (31 neighbors) |       | +0.72   | +1.40  | N/A     | N/A   | N/A   | -2          | -50   | +261  | +203  |

Nomenclature: See separate page